



Praveen
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AI SYSTEMS ARCHITECT

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📍 Taipei, Taiwan

CORE EXPERTISE

AI Architecture

AI-native workflows · agent orchestration
· multi-agent · RAG · domain-driven design

LLM Engineering

LangGraph · DSPy · prompt design · structured
output · LLM-as-judge evals · distillation

Vision & ML

VLMs (Gemini Vision) · object detection · GNN
· VAE · PyTorch · TensorFlow

Optimisation & Infra

quantisation · TensorRT · CUDA · OpenVINO
· 8-GPU Kubernetes · AWS · Docker

Security & Safety

OWASP LLM Top 10 · AI ethics gating · SAST /
AST · WCAG 2.1 AA

Leadership

platform ownership · evaluation frameworks
· code-review standards · mentoring

STACK

Python · TypeScript · FastAPI · Next.js 15
· Pydantic v2 · PostgreSQL / pgvector · Dagster
· Celery / Redis · SQLAlchemy 2.x

SELECTED IMPACT

1,300+ platform users, sole architect

2× faster distilled model at parity

15× HPC speedup, published at IEEE

8-GPU distributed training cluster

PROFILE

Senior AI Systems Architect with six years shipping production AI end to end, from blank repo to org-wide platforms. I own the full stack, LLM and agent orchestration, schema-first APIs, retrieval, security and deployment, and set the architecture principles and engineering standards that teams build against. Recently sole architect of a CEO-office enterprise AI platform serving 1,300+ users, turning company strategy into AI-native workflows. Deep range across vision-language models, model optimisation and distributed training. Guided by one rule: *prompts for judgment, code for determinism*.

EXPERIENCE

Senior AI Systems Architect

Mar 2022 – Present

ViewSonic · CEO Office | Taipei, Taiwan

- **Sole architect of an enterprise AI platform** (a strategy-execution OS), blank repo to 1,300+ users: converts company strategy and org data into AI-native workflows for goal-setting, evaluation and workforce AI-maturity. Own every layer, from model selection to UI.
- **AI-native workflow architecture.** Designed a reusable pattern, conversational elicitation → structured scoring → deterministic aggregation → narrative generation, using DSPy signatures with behavioural anchors and hysteresis to remove score flicker; set the principle *prompts for judgment, code for determinism*.
- **Full-stack delivery.** Next.js 15 and FastAPI on a domain-driven design, Dagster orchestration, PostgreSQL/pgvector RAG, SQL Server ingestion and SSE streaming, built to WCAG 2.1 AA.
- **Security & AI-safety lead.** Treated internal AI as an external-facing product: OWASP middleware, rate limiting, CSP/HSTS, SQL-injection AST validation, ethics-policy versioning with forced re-acknowledgment and per-turn safety flags; full OWASP LLM Top 10 compliance.
- **Evaluation frameworks.** Encoded Doerr-style OKR criteria into LLM-validated rubrics and an L1–L5 competency model with behavioural anchors, moving delivery from “feature shipped” to measured quality.
- **Earlier impact.** Built a six-layer vision-language document pipeline (Gemini Vision → Pydantic schema → AI-auditor alignment loop) for brand-compliant output; distilled a 600M-parameter teacher into a 2× faster production student at parity; ran an 8-GPU Kubernetes cluster for distributed PyTorch; object detection at mAP 0.82 over 500K images.
- **Technical leadership.** Set engineering standards and mentored engineers through multi-layer code review (local, pre-PR, AI-bot GitHub workflows) and process automation.

AI Software Developer

Aug 2020 – Mar 2022

Neurelli | Taipei, Taiwan

- Shipped industrial computer vision to production: licence-plate recognition accelerated with TensorRT/CUDA, and LCD-defect detection via a variational autoencoder, running the full cycle (data curation, training, optimisation, containerised deployment) at 85% production accuracy.

EDUCATION

M.Sc. Electrical Engineering & Computer Science

2018 – 2020

National Taipei University of Technology | Taiwan

- Thesis: GPU acceleration of flood simulation with high-performance computing and CUDA, achieving a 15× speedup; published at IEEE.